NuPa category reorganization

Discussion draft | 28 October 2026
Full audit of the former Collider Physics section

Audit scope. All **123 retained papers**: titles and INSPIRE abstracts checked individually, with full-text sections checked for category boundaries and mixed systems. This is a classification audit, not a scientific validity review.

Comparison. Against the pre-trial README at commit **6d4eaf9**: **23 primary-category moves** and **2 scope exclusions** from the original 125 papers. The full audit changed memberships for 22 papers, including 7 additional primary moves. The subsequent shortening of category names makes no further primary moves. Renaming alone is excluded from the paper lists.

Names and primary paper counts

Parent: Collider Physics → Particle Scattering and Decays. Compton scattering and positronium decays extend the scope beyond collider experiments.

Previous subcategory	Revised subcategory	Before	After
Top Quark Sector	Top Quarks	29	29
Electroweak Bosons and Higgs	W, Z and Higgs Bosons	33	27
Flavor and Hadron Spectroscopy	Hadrons	36	32
New category	Charged Leptons	-	6
New category	Photons	-	10
Observables and Methodology	General Methods	27	19

Names and placement rules

Short system names. Top Quarks, W, Z and Higgs Bosons, Hadrons, Charged Leptons and Photons use particle names consistently. General Methods identifies contributions across systems.

Top and hadron formation. Top decays precede ordinary hadron formation. Lighter quarks are accessed through the hadrons they form; Hadrons also includes inference of quark spin from hadron polarization and fragmentation correlations.

One primary category. Use the central system or contribution. Tau-pair and photon-polarization studies go to their respective categories, including Higgs-decay samples. Higgs/scalar-sector parameter and symmetry studies stay with Higgs.

Substantive cross-lists. Add a category when its states or interactions are actually analyzed. An exchanged particle, decay analyzer, detector conversion, citation or collider name alone is insufficient. Mixed systems can warrant several categories.

General Methods. Process-independent frameworks, foundational inference studies and surveys across systems belong here. Reviews focused on one physical system stay with that system.

Status. A trial organization in README and category descriptions. Canonical paper-data synchronization and regeneration of the living-review PDF remain pending. The full 123-paper decision ledger accompanies this summary as category_classification_audit.json.

Primary moves: particle systems

Moves 1-8 | Prior names omit the common Collider Physics prefix

A. Tau-pair states → Charged Leptons (6)

Shared reason: the analyzed state is the tau pair, including its decay reconstruction and spin correlations. A Higgs decay can motivate a cross-list without determining the primary category.

1. Quantum tomography with τ leptons at the FCC-ee
arXiv:2405.09201 | INSPIRE 2787034

From: Electroweak Bosons and Higgs. **Cross-list:** W/Z/Higgs retained for the explicit weak-mixing-angle and neutral-current-coupling analysis.

2. Prospects of measuring quantum entanglement in ττ final states at a future e+e- Higgs factory
arXiv:2409.20239 | INSPIRE 2835996

From: Electroweak Bosons and Higgs

3. Probing entanglement and testing Bell inequality violation with e+e- → τ+τ- at Belle II
arXiv:2311.17555 | INSPIRE 2727866

From: Flavor and Hadron Spectroscopy

4. Experimental aspects of the Quantum Tomography of tau lepton pairs at a Higgs factory collider
arXiv:2603.12609 | INSPIRE 3129062

From: Flavor and Hadron Spectroscopy. **Cross-list removed:** W/Z/Higgs. “Higgs factory” denotes the collider setting; the analysis concerns continuum tau-pair reconstruction.

5. Testing Bell inequality through h→ττ at CEPC
arXiv:2309.08103 | INSPIRE 2698185

From: Electroweak Bosons and Higgs. **Cross-list:** W/Z/Higgs retained for the Higgs-to-tau-pair channel.

6. Higgs Boson Spookiness: Probing Quantum Nonlocality with Spacetime-Resolved H→τ+τ- Decays
arXiv:2603.28868 | INSPIRE 3137624

From: Electroweak Bosons and Higgs. **Cross-list:** W/Z/Higgs retained for the spacetime-resolved Higgs-to-tau-pair channel.

B. Correct the actual system (2)

7. Quantum Information in High-Energy Physics: a top subject
arXiv:2610.00467 | INSPIRE 3210446

Observables and Methodology → Top Quarks. A focused review of top-pair states and their tomography, not a survey across systems.

8. Odd Physics Off the Diagonal: Constraining CP-violating SMEFT with Quantum Tomography
arXiv:2604.21857 | INSPIRE 3148517

Top Quark Sector → W, Z and Higgs Bosons. The worked process is WZ production and its spin density matrix. No Top cross-list is retained.

Primary moves: general methods

Moves 9-13 | Plus the two exclusions already agreed

C. General contributions across systems (5)

9. Quantum state tomography, entanglement detection and Bell violation prospects in weak decays of massive particles
arXiv:2209.13990 | INSPIRE 2157995

From: Electroweak Bosons and Higgs. A general spin-reconstruction framework with explicit top, top-W, weak-boson and Higgs-decay applications. **Cross-lists:** Top and W/Z/Higgs.

10. SMEFT Constraints via Quantum Entanglement in Portal-Mediated Scattering
arXiv:2610.09751 | INSPIRE 3212551

From: Electroweak Bosons and Higgs. A general SMEFT interference and entropy framework. **Cross-list:** W/Z/Higgs for the representative gauge-portal operator; quantitative channel phenomenology is deferred.

11. Maximal Entanglement in High Energy Physics
arXiv:1703.02989 | INSPIRE 1516805

From: Electroweak Bosons and Higgs. A common entanglement principle is applied to QED reactions and weak mixing. **Cross-lists:** Charged Lepton, Photon and W/Z/Higgs.

12. An open--quantum--system formulation of particle decay
arXiv:quant-ph/0602116 | INSPIRE 729754

From: Flavor and Hadron Spectroscopy. A general formulation of unstable-state evolution, with particle decay incorporated into a probability-conserving Lindblad description.

13. Flavor Patterns of Fundamental Particles from Quantum Entanglement?
arXiv:2410.23343 | INSPIRE 2844264

From: Flavor and Hadron Spectroscopy. A shared principle is applied to CKM and PMNS mixing. **Cross-lists:** Hadrons and Charged Lepton. The former boson cross-list is removed; W/Z exchange alone is insufficient.

Excluded from the review scope (2)

Shared reason: discrete-symmetry tests using angular correlations, with no central QIS analysis. In R2, “tomography” refers to event imaging and spin-axis estimation, rather than quantum-state reconstruction.

R1. Potential of the J-PET detector for studies of discrete symmetries in decays of positronium atom - a purely leptonic system
arXiv:1602.05226 | INSPIRE 1422075

R2. Testing CPT symmetry in ortho-positronium decays with positronium annihilation tomography
arXiv:2112.04235 | INSPIRE 1955404

Primary moves: photon systems

Observables and Methodology → Photons | Moves 14-23

Shared reason: photon polarization, production, annihilation or scattering in specified physical systems is the central subject.

D. High-energy photon pairs and polarimetry (2)

14. Quantum entanglement of photon pairs at proton-proton colliders
arXiv:2607.00971 | INSPIRE 3176266

W/Z/Higgs cross-list retained: both Higgs-to-diphoton and continuum diphoton sensitivities are explicitly calculated.

15. High Energy Photon Polarimetry at Lepton Colliders: Quantum Information from Converted Photons
arXiv:2607.08816 | INSPIRE 3179608

E. Positronium annihilation photons (1)

16. Genuine Multipartite Entanglement in the 3-Photon Decay of Positronium
arXiv:1706.06505 | INSPIRE 1606137

F. Compton-scattering systems (7)

17. Witnessing Entanglement In Compton Scattering Processes Via Mutually Unbiased Bases
arXiv:1807.04934 | INSPIRE 1682310

18. Probing entanglement in Compton interactions
arXiv:1910.05537 | INSPIRE 1758833

19. Entanglement Entropy of Compton Scattering with a Witness
arXiv:2305.10027 | INSPIRE 2660572

Charged Lepton cross-list added: final-electron and witness-photon polarization mutual information is explicitly calculated.

20. Measuring the Evolution of Entanglement in Compton Scattering
arXiv:2406.14352 | INSPIRE 2800856

21. Compton scattering mediated by quantum entanglement
DOI: 10.1103/PhysRevA.111.053708 | INSPIRE 2919423

22. Entanglement, steering, and separability in Compton-scattered annihilation photons
DOI: 10.1103/j7p7-ckp9 | INSPIRE 2968534

23. Photon-nucleon entanglement in Compton scattering at low and high energies
arXiv:2608.05330 | INSPIRE 3188079

Hadrons and Charged Lepton cross-lists added: nucleon polarizability effects and polarized photon-electron states are explicitly analyzed.

A legitimate photon / weak-boson overlap

C1. Exploring Bell inequalities and quantum entanglement in vector boson scattering
arXiv:2306.17247 | INSPIRE 2673433

W/Z/Higgs primary retained; Photon cross-list added. Separate photon-photon and photon-W polarization analyses accompany the W/Z scattering channels.

Cross-lists: physical systems

These entries retain their primary category; they are additional to moves 1-23.

C2. Constraining new physics in entangled two-qubit systems: top-quark, tau-lepton and photon pairs
arXiv:2208.11723 | INSPIRE 2141744

Top primary. Charged Lepton and Photon cross-lists added; the old Hadrons cross-list removed. Separate top, tau and photon systems are analyzed. W/Z/Higgs is retained for Higgs decays to taus and photons; the EFT cross-list is unchanged.

C3. Entanglement and Bell nonlocality with bottom-quark pairs at hadron colliders
arXiv:2406.04402 | INSPIRE 2796017

Hadrons primary; Top cross-list removed. The study concerns bottom pairs. Earlier top results supply motivation, without a separate top analysis.

C4. Full quantum tomography of top quark decays
arXiv:2402.14725 | INSPIRE 2760765

Top primary; W/Z/Higgs and Hadrons cross-lists added. W spin and bottom spin are explicit parts of the analyzed orbital-W-bottom state, with separate correlations calculated.

C5. Post-decay quantum entanglement in top pair production
arXiv:2307.06991 | INSPIRE 2677292

Top primary; W/Z/Higgs cross-list added. The paper calculates post-decay top-W and W-W states.

C6. Entanglement autodistillation from particle decays
arXiv:2401.06854 | INSPIRE 2747189

General Methods primary; Top retained and W/Z/Higgs added. The general decay framework is applied to the top-W state and its correlation increase.

C7. Entanglement Maximization and Symmetry Selection in Composite Higgs Models
arXiv:2605.17434 | INSPIRE 3157227

W/Z/Higgs primary; Top cross-list added. Higgs-sector symmetry and potential are the objective; the explicit calculation analyzes the top-pair helicity state in Higgs scattering.

C8. Quantum tomography for collider physics: Illustrations with lepton pair production
arXiv:1707.01638 | INSPIRE 1608880

General Methods primary; W/Z/Higgs retained. Dilepton angles probe an intermediate production/polarization state. Final lepton spins are summed over, so the interim Lepton cross-list is removed.

Cross-lists: methods and surveys

Specific applications are exposed without changing the general primary category.

C9. Quantum entanglement and Bell inequality violation at colliders
arXiv:2402.07972 | INSPIRE 2757981

Add Top, Hadrons, Charged Lepton and Photon; retain W/Z/Higgs. Dedicated sections cover all five systems, including mesons, hyperons, tau pairs and Higgs-to-diphoton states.

C10. Tests of quantum contextuality in particle physics
arXiv:2504.12382 | INSPIRE 2913431

Add Hadrons and Charged Lepton; retain Top and W/Z/Higgs. The paper contains meson/baryon analyses and a tau-pair feasibility study, in addition to top and weak-boson tests.

C11. Quantum Information meets High-Energy Physics: Input to the update of the European Strategy for Particle Physics
arXiv:2504.00086 | INSPIRE 2906652

Add Hadrons and Charged Lepton; retain Top and W/Z/Higgs. The strategy paper develops B-meson and tau-pair programs. Introductory photon examples do not warrant a Photon cross-list.

C12. Colliders are not Testing Locality via Bell's Inequality nor Providing an Unconditional Proof of Entanglement
arXiv:2507.15949 | INSPIRE 2952954

Add Top and Charged Lepton; retain W/Z/Higgs. The general critique explicitly addresses top-pair, tau-pair and Higgs-to-vector-boson proposals.

C13. A General Analysis for Observing Quantum Interference at Colliders
arXiv:2201.03159 | INSPIRE 2006954

Add W/Z/Higgs and Photon. A worked diphoton example compares interference of scalar and spin-two Higgs amplitudes within the general framework.

C14. Quantum Tomography Measures Entanglement in Collider Reactions
arXiv:1910.06311 | INSPIRE 1759762

Add W/Z/Higgs. The general discussion explicitly uses a reconstructed Z-production polarization matrix.

C15. Entanglement Detection with Rotationally Covariant Measurements -- From Compton Scattering to Lemonade
arXiv:2512.03984 | INSPIRE 3089176

Photon cross-list retained. A general measurement framework has explicit Compton-scattering and polarized-light applications.